

Transient Rotor Angle Stability Prediction Based on Deep Belief Network and Long Short-term Memory Network

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Abstract: In this paper, a transient stability prediction method based on deep belief network (DBN) and long short-term memory (LSTM) network is proposed. DBN is utilized to embed the original transient stability data into a low-dimensional representation space and assess transient stability preliminarily. Owing to the good performance of LSTM networks in extracting the features of time series for a long-time span, it is utilized to predict generator rotor angle trajectory of instability samples, which can identify the unstable generator in advance. The proposed method is validated by IEEE New England 39-bus system. Simulation results show that the proposed method has the merits of quickness and accuracy criteria and can provide guidelines for online monitoring of power system stability.

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Keywords: Transient stability assessment, DBN, LSTM, rotor angle stability.

1. INTRODUCTION

With rapid growth of electricity demand and increasing uncertainties, power system often operates quite close to its stability boundary nowadays, which requires online safety and stability analysis of power systems to be more reliable and accurate. Besides, recent severe blackouts have demonstrated the importance and need of transient stability prediction (Carreras et al., 2016). If generator rotor angle trajectory can be predicted quickly and accurately, the control measures can be taken in advance to maintain power system stable. Therefore, online power system stability assessment and real-time prediction technology of disturbed trajectories have received extensive attention.

The prevalent power system transient stability analysis methods mainly include time domain simulation (Kundur, 1994), direct methods (including the Lyapunov method and Transient Energy Function), (Chiang, 2010), Extended Equal Area Criteria (EEAC) (Xue et al., 1989) and Artificial Intelligence (AI). Time domain simulation method is detailed and accurate, but its disadvantage is that the calculation amount is too large, and the online information of power system stability cannot be provided. The accuracy of direct methods is not high, and it has some challenges and limitations involved in the practical applications for power system transient stability assessment. EEAC method can find the stability margin of the system, but its online computing ability needs further development. These methods are currently not fully compatible with the actual requirements of online calculations.

With the development of wide area measurement system (WAMS) and phasor measurement units (PMUs), various system dynamic information can be measured, which contains critical information for assessing the system security. Generator active power, reactive power, rotor angle and the dynamic response trajectories following a disturbance can be directly measured.

AI method can find the relationship between these real-time data and power system transient stability, and it can quickly establish a nonlinear mapping relationship between input data and output data. There are many AI methods used in power system transient stability assessment, such as artificial neural network (ANN) (Al-Masri et al., 2013), decision tree (DT) (Zhu et al., 2016), support vector machine (SVM) (You et al., 2013) and so on. Unfortunately, these shallow learning networks have limited ability to extract input features. Therefore, deep learning models provide a new idea for power system transient stability prediction.

The transient rotor angle trajectory of power system can provide system stability information intuitively and concisely. At present, the disturbed rotor angle trajectory prediction technology mainly includes model-based method and model-free method. Model-based method (Wang et al., 2012) is difficult for real-time transient stability analysis, due to its dependence on the scale of power system. Model-free method mainly includes curve fitting and extrapolation method (Huang et al. 2018) and AI method (Zhou et al., 2016). Although the trend extrapolation method is entirely based on numerical sequence analysis and data mining, it is difficult to find a suitable model to predict the rotor angle curve of each generator after fault. The Recurrent Neural Network (RNN),

which is one of the deep learning algorithms, can extract time series features. LSTM is an improved RNN that can better fit longer time series and predict them accurately. Now, using machine learning to predict rotor angle trajectory is in the initial stage.

In this paper, a prediction model based on DBN and LSTM is proposed. The model firstly uses the DBN classifier to assess the transient rotor angle stability of the power system, and then uses the LSTM network to predict the generator rotor angle trajectory for unstable samples. This method can not only predict the transient rotor angle stability of the power system, but also predict the trend of the generator rotor angle trajectory in advance. It visually finds the unstable generator without complicated calculations; thus it can provide necessary information for the system stability control measures.

2. THEORY

2.1 Deep Belief Network

As a multi-layer Restricted Boltzmann Machines (RBM) model, DBN is a deep learning model, whose network structure is shown in the Fig. 1. When applying DBN for transient stability assessment, the network structure should be trained first to determine the connection weights and biases. The training process consists of two parts: pre-training and fine-tune. Pre-training process uses an unsupervised greedy algorithm to train each layer of RBM separately, and to ensure that feature information is reserved as much as possible when mapping the feature vector to the higher-layer (Nair et al., 2010). Pre-training process provides a good initial weight value for the entire network. Then, DBN uses the traditional back propagation (BP) algorithm to fine tune the parameters of RBM model to make the model converge to the best.

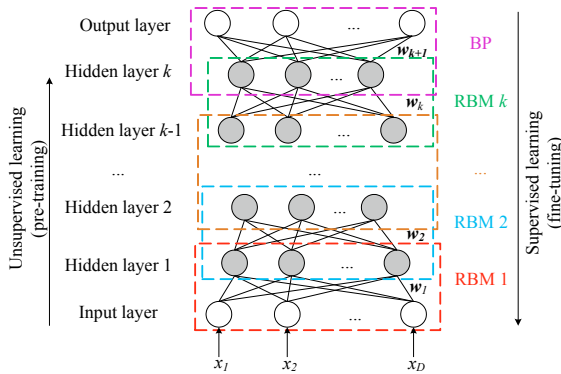


Fig. 1 The structure of DBN model

An RBM consists of a visible layer $\mathbf{v}$ and a hidden layer $\mathbf{h}$. Visible layer connects to the observation data. Hidden layer is used to extract the effective characteristics of the input data. The units in the same layer are not connected, and the visible units and hidden units are connected by weights $\mathbf{w}$. The states of unit in the network are only inactive and active, usually represented by binary 0 and 1. A joint configuration, $(\mathbf{v}, \mathbf{h})$ of

the visible and hidden units has an energy given by (Hinton et al. 2006)

$$E(\mathbf{v}, \mathbf{h}|\boldsymbol{\theta}) = -\sum_{i=1}^n a_i v_i - \sum_{j=1}^m b_j h_j - \sum_{i=1}^n \sum_{j=1}^m v_i w_{ij} h_j \quad (1)$$

where $\boldsymbol{\theta} = (w_{ij}, a_i, b_j)$ is a parameter of RBM, v_i, h_j are the binary states of visible unit i and hidden unit j , a_i, b_j are their biases and w_{ij} is the weight between them, n, m are the number of visible units and hidden units. Based on the energy function, the joint probability distribution is given by

$$p(\mathbf{v}, \mathbf{h}|\boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} \exp(-E(\mathbf{v}, \mathbf{h}|\boldsymbol{\theta})) \quad (2)$$

where $Z(\boldsymbol{\theta}) = \sum_{\mathbf{v}, \mathbf{h}} \exp(-E(\mathbf{v}, \mathbf{h}|\boldsymbol{\theta}))$ is a normalizing constant.

2.2 Input Feature Set of DBN Classifier

In the analysis of transient rotor angle stability using DBN, the key is the construction of feature set. The relevance extent between the selected input data and the transient stability of the power system will greatly affect the clustering ability and the classification ability of the DBN classifier.

As for single-machine infinite bus system, generator rotor swing equations is shown below

$$\begin{cases} \frac{d\delta}{dt} = \omega_0 (\omega - 1) \\ M_j \frac{d\omega}{dt} = P_m - D(\omega - 1) - P_e \end{cases} \quad (3)$$

where $M_j, D, P_m, P_e, \delta, \omega_0, \omega$ are inertia constant, damping coefficient, mechanical power, electrical power, rotor angle of machine, synchronous angular velocity, angular velocity in p.u. When a serious disturbance occurs in the power system, the input mechanical power and the output electromagnetic power of generator will be out of balance, causing generator rotor swing and the angular velocity and rotor angle change. The angular velocity and acceleration can characterize the instability tendency of the system.

From the perspective of system energy, the power system transient rotor angle stability essentially reflects the problem of whether the unbalanced energy injected during the fault can be absorbed. The rotor motion equation after the fault is

$$\frac{M_j d^2 \delta}{\omega_0 dt^2} = P_m - P_e \quad (4)$$

where the right side of the equation (4) is the acceleration power of the generator after the fault. Through integration, equation (4) becomes

$$\begin{aligned} \int_{\delta_0}^{\delta_c} (P_m - P_e) d\delta &= \int_{\delta_0}^{\delta_c} \frac{M_j}{\omega_0} \dot{\delta} d\dot{\delta} \\ &= \frac{1}{2} \frac{M_j}{\omega_0} (\dot{\delta}_c^2 - \dot{\delta}_0^2) \end{aligned} \quad (5)$$

where δ_0 and δ_c indicate the generator rotor angle value when the fault occurs and when the fault is cleared, $\dot{\delta}_0$ and $\dot{\delta}_c$ indicate the generator angular velocity difference when the fault occurs and when the fault is cleared. The left side of equation (5) corresponds to the acceleration area, and the right side represents the increase of rotor kinetic energy during relative motion. When the deceleration area is not bigger than the acceleration area, the system is transiently stable. It can be seen from equation (5) that the difference in angular velocity and the change of rotor kinetic energy can characterize the disturbance effect on the system.

After a multi-machine system is disturbed, the performance of the rotor angle instability is that the angular velocity of any two generators is different and the difference of the rotor angle is larger and larger. Therefore, the system stability can be reflected by calculating the rotor angle difference. Analogous to a single-machine infinite bus system, the rotor kinetic energy, rotor angle value, angular velocity, angular acceleration, system load level, mechanical power and electrical power are considered as input features.

Based on the existing literatures (Tso et al., 1998; Ferreira et al., 2016), this paper constructs a feature set which is not affected by the system scale, reflecting the operating state of the power system, to assess transient stability. The feature set is shown in table 1, and t_0 is the fault occurrence time and t_c is the fault clearing time.

Table 1. Input feature set

No.	Feature description
1	Load level of system
2	Total mechanical power before t_0
3	Maximum active power impact at t_0
4	Minimum active power impact at t_0
5	Sum of generator acceleration power at t_0
6	Maximum generator relative angular acceleration at t_0
7	Minimum generator relative angular acceleration at t_0
8	Rotor angle of generator with max. acceleration at t_0
9	Difference of max. and min. generator rotor angle at t_0
10	Difference of max. and min. generator angular velocity at t_0
11	Difference of max. and min. generator angular acceleration at t_0
12	Difference of max. and min. generator rotor kinetic energy at t_0
13	Maximum generator rotor kinetic energy at t_0
14	Difference of max. and min. generator rotor angle at t_c
15	Difference of max. and min. generator angular velocity at t_c
16	Difference of max. and min. generator angular acceleration at t_c
17	Maximum generator rotor kinetic energy at t_c
18	Kinetic energy of generator with max. rotor angle at t_c
19	Rotor angle of generator with max. kinetic energy at t_c
20	Maximum rotor angle variation of all generators at t_c
21	Total energy adjustment of system

Feature (1, 2) reflect the basic system operating conditions. The system with high load level often suffers the larger risk of instability. When the fault occurs, each generator first shares the impact power according to the electric distance from the fault point. Feature (3, 4) can essentially reflect the electric distance from the generator to the fault point. Feature (5) can reflect the power imbalance of all generators and the rotor acceleration. Feature (8, 9, 14, 19, 20) related to the rotor angle reflect the synchronization relationship of generators in power system. Feature (12, 13, 16-18, 21) related to system energy characterize the magnitude of the fault impact on the system. Both feature (10, 15) related to the angular velocity and feature (6, 7, 11) related to angular acceleration characterize the motion and disturbance of the rotor, which can reflect the stability trend of the system.

2.3 Long Short-Term Memory Unit

RNN is an improved class of artificial neural network using the temporal information of the input data (Graves, 2012). The neurons in the same hidden layer have inner connection so that they can transmit the temporal information. Unfortunately, one of problems of RNNs is the vanishing gradient which is that the influence of a given input on the hidden layer and therefore on the network output, either decays or blows up exponentially as it cycles around the network's recurrent connections.

LSTM network can address the vanishing gradient problem by using LSTM memory blocks to take place of hidden units in RNN. A LSTM unit consists of a cell, an input gate, an output gate and a forget gate (Hochreiter et al., 1997). The cell remembers values over arbitrary time intervals and the three gates regulate the flow of information into and out of the cell. A single LSTM unit structure used in this paper is illustrated in Fig. 2 and equation (6)-(10).

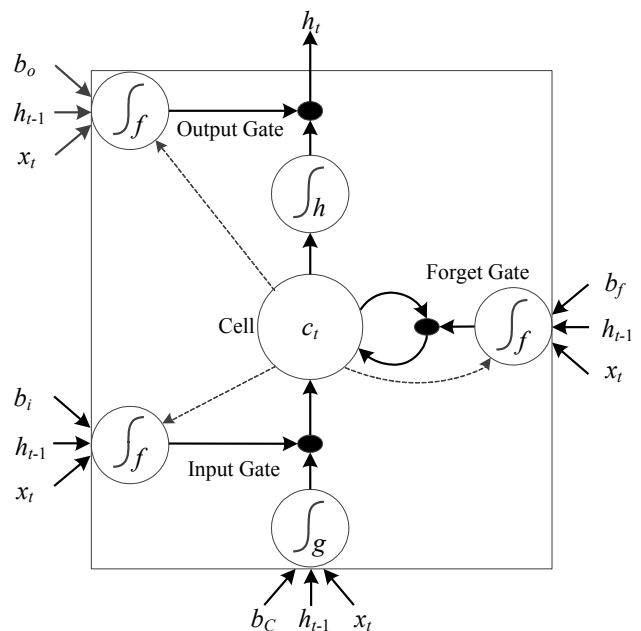


Fig. 2 The structure of LSTM unit

$$i_t = \sigma(W_i h_{t-1} + W_i x_t + W_i c_{t-1} + b_i) \quad (6)$$

$$f_t = \sigma(W_f h_{t-1} + W_f x_t + W_f c_{t-1} + b_f) \quad (7)$$

$$c_t = f_t c_{t-1} + i_t \tanh(W_c h_{t-1} + W_c x_t + b_c) \quad (8)$$

$$o_t = \sigma(W_o h_{t-1} + W_o x_t + W_o c_t + b_o) \quad (9)$$

$$h_t = o_t \tanh(c_t) \quad (10)$$

where σ is the logistic sigmoid function, and i_t, f_t, o_t and c_t are respectively the input gate, forget gate, output gate and the memory cell, all of which are the same size as the output vector h_t . W and b are weights and biases terms of the neurons, which need to be learned through gradient-based back propagation algorithm.

3. PREDICTION MODEL

3.1 Prediction Procedure

The transient rotor angle stability predictor consists of offline training and online implementation, whose flow chart is shown in Fig. 3.

3.1.1 Offline Training

Considering different operation scenarios of power system, the sample data of the power system can be generated by the time domain simulations, and the sample set of the DBN classifier and LSTM network can be built. Then, DBN and LSTM will be trained. A sub-training set is established for each generator. The parameters of LSTM are changed with supervision based on Tensorflow.

3.1.2 Online Implementation

If a fault occurs in the power system, online data can be obtained from the WAMS and the original input features can be constructed. The original input features are used as the input of the DBN for assessing the power system transient rotor angle stability. If the system is stable, the data will be updated for the next evaluation. If the system is unstable, the m steps of the generator rotor angle data after the fault will be sent to the LSTM. The LSTM model will predict the rotor angle trajectory of the next n steps, which can observe the out-of-step generator. If no out-of-step generator is observed, the system can be considered in a stable state.

3.2 Model Performance Index

To quantify the accuracy of the LSTM model, mean absolute percentage error (MAPE) and root mean square error (RMSE) are chosen as the evaluation index. The calculation formulas are given by

$$\delta_{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{y_i} \times 100\% \quad (11)$$

$$\delta_{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (12)$$

where n is predicted time step, and y_i is the true value of the rotor angle trajectory; $\hat{y}_i$ is the predicted value of the rotor angle trajectory. The smaller the value of MAPE, the higher the prediction accuracy. The value of RMSE reflects the actual situation of the error.

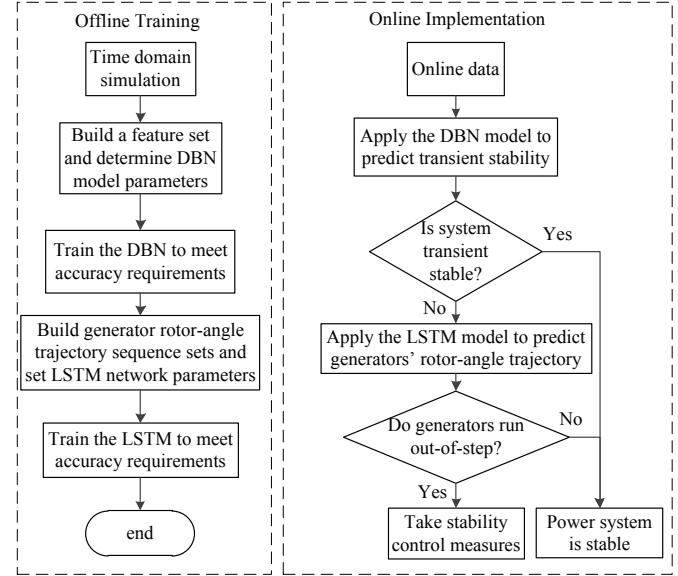


Fig. 3 Flow chart of transient rotor angle stability prediction

4. CASE STUDY

4.1 Test system

The proposed prediction algorithm is tested in the IEEE New England 39-bus system and the database is generated by using the DiGSILENT/PowerFactory. To construct a relatively complete sample space, the Latin hypercube sampling method is adopted. The load of the system is assumed to follow the normal distribution. The original load is the expected value and the variance is 0.08. The output of generator changes according to the load level, and the No. 2 generator is selected as the reference machine. The contingencies considered are mainly on each transmission line and fault locations are at 0%, 20%, 50%, and 80% of the length. The simulation assumes that the three-phase short-circuit occurs at 1s and is cleared at 1.2s. The simulation duration is 5s. At the end of the simulation, if the difference of any two generators rotor angles exceeds 360° (Wang et al., 2016), the system is considered as transient instability. In this paper, a total of 13,600 samples are obtained, of which 10,600 are used as training set and 3,000 are used as test set. The training set is used for training, and the test set is used to verify the validity of the model.

4.2 Result Analysis

4.2.1 Result of DBN Classification

The sample sets are firstly normalized. A three hidden-layer DBN network structure is established, which consists of RBM1, RBM2 and RBM3. The structural parameters are 24-20-15-10-1. The neurons number of visible layers, hidden layer of RBM1, RBM2, RBM3 and output layer is 24, 20, 15, 10 and 1. The learning rate and momentum of RBM are set to 0.5 and iteration is 500 times. In the stability evaluation, the number of hidden layer neurons in BP model is 30 and learning rate is 0.001. SVM parameters are determined by 10-fold cross validation. The classification accuracy and errors are shown in Table 2, where false dismissal (FD) means that the unstable case is classified as stable and false alarm (FA) denotes that the stable case is deemed as unstable. It can be seen from Table 2 that the DBN has the high accuracy and the small error, and it has a better performance than the BP and the SVM model.

Table 2. The predicted results of different models

Model	Accuracy rate/%	FD rate/%	FA rate/%
BP	96.60	1.30	2.10
SVM	97.23	1.13	1.63
DBN	97.80	1.30	0.90

4.2.2 Result of LSTM Rotor Angle Trajectory Prediction

For unstable samples classified by DBN, further rotor angle trajectory prediction is needed to identify the system out-of-step generator operating condition and take further control measures.

The rotor angle trajectory data generated by time domain simulation is classified according to different generators. Each generator corresponds to one LSTM model and the input time step after the fault is collected (this paper takes 20-time steps and one time step is 0.01s). The corresponding rotor angle data is used as the input of the model, and the rotor angle data of the next 200-time steps are respectively selected as the output of the model. 10,600 samples are taken as the training set, and one sample is selected as the test sample from the remaining data each time. When the DBN predicts that the system transient rotor angle is unstable, the rotor angle data of each generator collected by the WAMS is used as the input of the LSTM network to predict.

In this case, the three-phase short-circuit fault occurs at 1s, and is cleared at 1.2s. Fault location is on 50% of the line 26-29. The generator output and load level are set to 96%. For the generator which exceeds the limit, the output is set to the corresponding upper limit value. The 20-time steps of rotor angle values from 1s to 1.2s are collected as the input of the LSTM network to predict trajectory of next 200-time steps. The prediction errors of all generators are shown in Table 3. The prediction results of nine generators' rotor angle trajectories are shown in Fig. 4.

Table 3. The prediction results

Generator No.	MAPE/%	RMSE/°
G01	1.697	1.662
G02	0.000	0.000
G03	0.659	0.286
G04	1.806	0.739
G05	0.570	0.344
G06	1.404	0.573
G07	0.971	0.458
G08	1.281	0.498
G09	2.071	22.804
G10	0.868	0.802

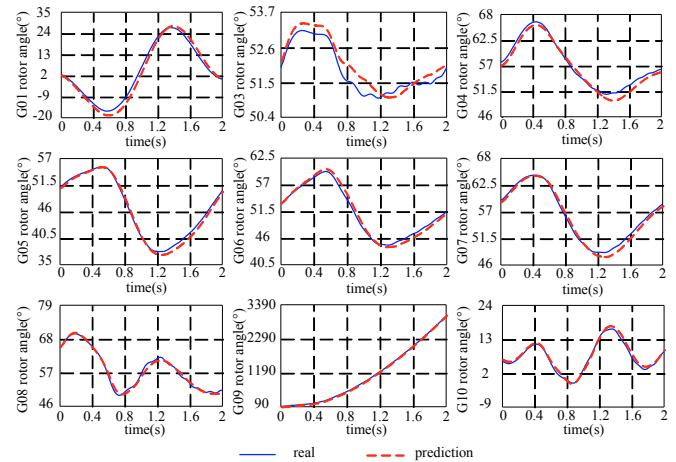


Fig. 4 Results of generator rotor angle trajectory prediction

It can be seen from Table 3 and Fig. 4 that when predicting next 1s rotor trajectory, MAPE is within 2.1%. The RMSE maximum is within 1.662° for stable generators and 22.804° for unstable generator. Prediction error is within the acceptable range and the predicted rotor angle trajectory trend is the same as the actual situation. The LSTM rotor angle trajectory predictor has high accuracy and can directly reflect unstable generator.

4.2.3 Result of Online Rotor Angle Trajectory Prediction

In this case, the three-phase short-circuit fault occurs at 1s, and location is on 50% of the line 26-27, and is cleared at 1.2s. The generator output and load level are set to 106%. For the generator exceeding the limit, the output is set to the corresponding upper limit value. DBN classifier assesses the system is unstable and then LSTM is used to predict its rotor angles. 20-time steps rotor angle values from 1s to 1.2s are collected as the input of the LSTM network. The prediction starts at 1.2s, and the prediction stage duration is 10ms. The prediction results of four generators are shown in Fig. 5.

It can be seen from Fig. 5, the result of the No. 9 generator out-of-step operation is obtained at 1.21s before the generator loses synchronism. LSTM rotor angle trajectory predictor can accurately predict the generator rotor angle trajectory and find the unstable generator in advance.

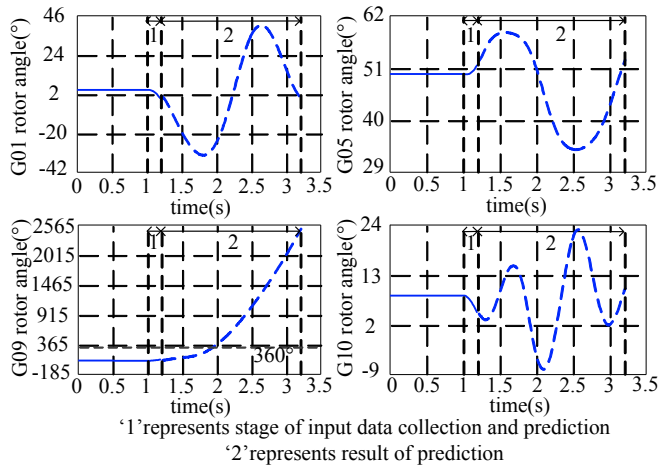


Fig. 5 Prediction curves of rotor angle trajectory

5. CONCLUSIONS

This paper proposed a transient rotor angle stability prediction model based on DBN and LSTM, which adds online prediction of generator rotor angle trajectory after DBN classifier. Compared with conventional machine learning methods used in power system stability assessment, the proposed method combines deep learning technology to find the law of nonlinear data and quickly predict the trajectory curves. The results show that DBN-LSTM model can assess power system transient stability and predict the rotor angle trajectory of each generator accurately. The prediction result can assist the dispatcher to take corresponding measures in the actual dispatch operation.

6. ACKNOWLEDGMNETS

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